

Lecture 1: Option Pricing & A Two-State Tree Model

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Based on lecture notes by me and many previous CS335 instructors

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Definition: European Put/Call Options

By purchasing a **European option**, the holder has the *right*, but *not obligation* to, at some specified time T in the future...

- Call option: buy one asset at a prescribed price K .
- Put option: Sell one asset at a prescribed price K .

T is called the expiry / exercise time.

K is called the strike price.

(We will generally assume the underlying asset is a stock.)

Exercising an Option

Let $S(t)$ be the evolving, time-dependent price of the asset, where t is time. It is referred to as *risky*, since its price changes unpredictably.

At expiry time $t = T$, the value V of the option (called the **payoff**) is:

- Call option: $\text{payoff} = \max(S - K, 0)$
- Put option: $\text{payoff} = \max(K - S, 0)$

Why is this true? ¹?

¹For an interactive visualization see: <https://colab.research.google.com/drive/1v8gYkZDdM3gWdNY6tU1HhWI7wIhx4RR0?usp=sharing>

Understanding the Payoff

A *call option* allows us to buy a stock for K at time T . Two possibilities at T :

- If $S(T) > K$

option)

- ① We buy the asset for K (from the issuer of the
- ② Then sell the asset at current price $S(T)$
(on the stock market)
- ③ We make positive profit, $S(T) - K > 0$.

- Else $S(T) \leq K$

- ① Let the option expire, i.e., do nothing. (There is no benefit to buying it at the higher price K - We'd lose money!) Thus the value is 0.

Similar logic for European *put options* gives $\max(K - S(T), 0)$

We only use the option if it would benefit us,
thus the value/payoff of a European call option
at expiry is $\max(S - K, 0)$.

Option Payoff - Comments

In practice, options contracts are usually settled in cash *directly* instead— no actual buying/selling of assets occurs.

Definition:

- A European option can only be exercised at the
 - expiry time T .
- An American option may be exercised before T .

This leads to a key topic of this course...

Option Pricing

Central question: what is the value of the option at *other* times, $t < T$?
That is, what should the price of the option be *earlier*?

For example, when an option is initially sold, what should the seller charge?

This is the problem of option pricing. It is a core theme of this course. (We will use hedging to minimize or avoid risk, and allow us to price options.)

End of Lecture 1